DISASTER RESPONSE RESOURCE ALLOCATION SYSTEM

USE CASE SPECIFICATION REPORT

Prepared By	Pratibha Yadav (25MCMC05)
Date Created	September 2026
Document Type	Use Case Specification

1. Use Case Diagram

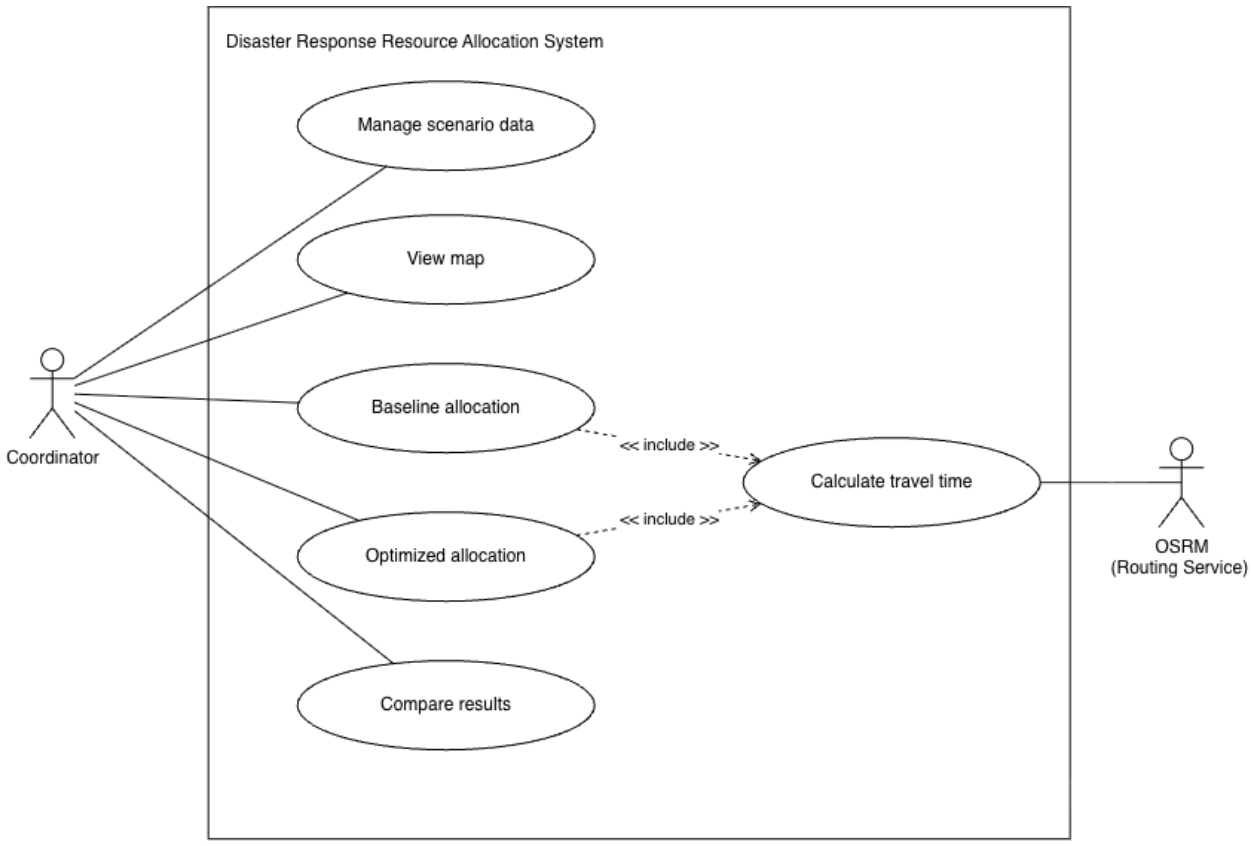


Figure 1. Disaster Response Resource Allocation System Use Case Diagram

Use Case Specifications

DR-001	Manage Scenario Data
Use Case ID	DR-001
Use Case Name	Manage Scenario Data
Created By	Pratibha Yadav , Garv Agrawal
Last Updated By	Pratibha Yadav , Garv Agrawal
Date Created	September 2026
Date Last Updated	September 2026
Actors	Coordinator
Description	This use case is used by the coordinator to create a disaster scenario and add the affected areas and resource bases needed for allocation.
Preconditions	None. This is an entry point of the system.
Post conditions	Scenario, area, and resource data are stored and available to the rest of the system.
Normal Flow	1. Coordinator creates a new scenario. 2. Coordinator adds affected areas with population and severity. 3. Coordinator adds resource bases with available resources. 4. System stores the entered data.
Alternate Flows	If required fields are missing, the system prompts the coordinator to complete them.
Exceptions	If the database is unavailable, the system shows an error and does not save the data.
Priority	High
Frequency of Use	Frequent
Business Rules	Severity must be one of Low, Medium, High, or Critical.
Assumptions	Resource and demand values used are simulated, not live operational data.

DR-002	View Map
Use Case ID	DR-002
Use Case Name	View Map
Created By	Pratibha Yadav , Garv Agrawal
Last Updated By	Pratibha Yadav , Garv Agrawal
Date Created	September 2026
Date Last Updated	September 2026
Actors	Coordinator
Description	This use case is used by the coordinator to see the affected areas and resource bases plotted on a map before running an allocation.

Preconditions	At least one area or resource base has been added (see Manage Scenario Data).
Post conditions	The coordinator can visually review the scenario.
Normal Flow	1. Coordinator opens the dashboard. 2. System loads the scenario's areas and bases. 3. System displays them on the map, with areas colored by severity.
Alternate Flows	If no data exists yet, the system shows an empty map with a message to add scenario data first.
Exceptions	If map tiles fail to load, the system shows a fallback list view of areas and bases.
Priority	Medium
Frequency of Use	Frequent
Business Rules	None.
Assumptions	OpenStreetMap tiles are reachable.

DR-003	**Run Baseline Allocation**
Use Case ID	DR-003
Use Case Name	Run Baseline Allocation
Created By	Pratibha Yadav , Garv Agrawal
Last Updated By	Pratibha Yadav , Garv Agrawal
Date Created	September 2026
Date Last Updated	September 2026
Actors	Coordinator
Description	This use case runs the simple nearest-base allocation method, used as a comparison baseline for the optimized method.
Preconditions	Scenario data (areas and resources) has already been entered.
Post conditions	The baseline allocation and its metrics are stored and available for viewing.
Normal Flow	1. Coordinator selects “Run Baseline”. 2. System invokes Calculate Travel Time. 3. System assigns each area's demand to the nearest base with available resources. 4. System calculates response time and unmet demand. 5. System stores and displays the result.
Alternate Flows	If the nearest base has run out of resources, the system moves to the next-nearest base with availability.
Exceptions	If travel-time data cannot be retrieved, the run fails and an error is shown.
Priority	High
Frequency of Use	Frequent
Business Rules	A resource base cannot send more resources than it currently has available.

Assumptions	None.

DR-004	**Run Optimized Allocation**
Use Case ID	DR-004
Use Case Name	Run Optimized Allocation
Created By	Pratibha Yadav , Garv Agrawal
Last Updated By	Pratibha Yadav , Garv Agrawal
Date Created	September 2026
Date Last Updated	September 2026
Actors	Coordinator
Description	This use case runs the optimization-based allocation, considering resource limits, demand, and travel cost together.
Preconditions	Scenario data has already been entered.
Post conditions	The optimized allocation and its metrics are stored separately from the baseline result.
Normal Flow	1. Coordinator selects “Run Optimized”. 2. System invokes Calculate Travel Time. 3. System builds and solves the optimization model under the defined constraints. 4. System stores and displays the result.
Alternate Flows	If the solver cannot find a feasible solution, the system reports which constraint could not be satisfied.
Exceptions	If the solver takes too long, an error is shown and the run is marked failed.
Priority	High
Frequency of Use	Frequent
Business Rules	Assigned quantities cannot be negative and cannot exceed base capacity or area demand.
Assumptions	An optimization solver (OR-Tools or PuLP) is available on the backend.

DR-005	**Compare Results**
Use Case ID	DR-005
Use Case Name	Compare Results
Created By	Pratibha Yadav , Garv Agrawal
Last Updated By	Pratibha Yadav , Garv Agrawal
Date Created	September 2026
Date Last Updated	September 2026

Actors	Coordinator
Description	This use case lets the coordinator compare the baseline and optimized allocation results side by side.
Preconditions	Both a baseline run and an optimized run exist for the selected scenario.
Post conditions	The coordinator can see and judge the difference between the two methods.
Normal Flow	1. Coordinator selects a scenario with completed runs. 2. Coordinator selects “Compare”. 3. System retrieves both results. 4. System displays the metrics and maps side by side.
Alternate Flows	If only one of the two runs exists, the system prompts the coordinator to run the missing method first.
Exceptions	None significant.
Priority	Medium
Frequency of Use	Occasional
Business Rules	None.
Assumptions	None.

DR-006	**Calculate Travel Time**
Use Case ID	DR-006
Use Case Name	Calculate Travel Time
Created By	Pratibha Yadav , Garv Agrawal
Last Updated By	Pratibha Yadav , Garv Agrawal
Date Created	September 2026
Date Last Updated	September 2026
Actors	System; OSRM (Routing Service)
Description	This use case is triggered automatically to calculate the travel time between each resource base and affected area. It is included by both the baseline and optimized allocation use cases.
Preconditions	The scenario's areas and bases already have location data.
Post conditions	A travel-time matrix is stored for use by the allocation use cases.
Normal Flow	1. System sends base and area coordinates to OSRM. 2. OSRM returns travel-time/route information. 3. System stores the travel-time matrix.
Alternate Flows	None.
Exceptions	If OSRM is unreachable, the system falls back to a straight-line distance estimate.
Priority	High
Frequency of Use	Automatic, on every allocation run

Business Rules	None.
Assumptions	OSRM and OpenStreetMap data are available and reasonably accurate.

DR-007	**Test Scenarios**
Use Case ID	DR-007
Use Case Name	Test Scenarios
Created By	Pratibha Yadav , Garv Agrawal
Last Updated By	Pratibha Yadav , Garv Agrawal
Date Created	September 2026
Date Last Updated	September 2026
Actors	Coordinator
Description	This use case is used to run the baseline and optimized allocation methods on more than one sample scenario, to check that the system works beyond a single case. It extends Run Baseline Allocation and Run Optimized Allocation.
Preconditions	More than one sample scenario has been created.
Post conditions	Results for each tested scenario are recorded for review.
Normal Flow	1. Coordinator creates 2–3 sample scenarios. 2. Coordinator runs baseline and optimized allocation for each scenario. 3. Coordinator records and reviews the results.
Alternate Flows	None.
Exceptions	If a scenario run fails, the coordinator is notified and can retry that scenario alone.
Priority	Low
Frequency of Use	Occasional, mainly during evaluation
Business Rules	None.
Assumptions	None.
